Hengameh Mirhajianmoghadam

Electrical and Computer Engineering | LinkedIn | Google Scholar | GitHub
Dissertation successfully defended | Ph.D. degree expected December 2026
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Research Interests

Autonomous aerial systems; guidance, navigation, and control; real-time model predictive control; event-based vision; visual odometry and SLAM; multisensor state estimation; distributed control and multi-agent coordination.

Education

Ph.D. in Electrical and Computer Engineering

New Mexico State University | January 2022–December 2026 expected

Dissertation successfully defended; degree conferral expected December 2026.

M.Sc. in Electrical and Computer Engineering

Ferdowsi University of Mashhad | September 2013–February 2016

Academic and Research Experience

Graduate Research Assistant — Autonomous Systems, Robotics, and Control

New Mexico State University | January 2022–Present

- Developed and experimentally validated model predictive control for fast trajectory tracking of commercial unmanned aircraft systems by combining system identification, constrained optimization, and real-time ROS implementation.
- Developed eRatSLAM by integrating event-based visual odometry, place recognition, and biologically inspired mapping to support localization and map formation using neuromorphic cameras.
- Developed event-based lane detection to estimate lane boundaries, lateral displacement, and heading error, providing visual feedback for closed-loop autonomous navigation.
- Integrated event-derived motion estimates with inertial measurements through Kalman filtering to improve state estimation for robotic navigation.
- Implemented event-based place recognition and loop-closure detection to identify revisited locations and maintain consistency in the SLAM map.
- Co-developed distributed model predictive control for multi-agent lattice formation and collision-aware coordination, evaluating performance against established coordination methods.
- Established experimental workflows for sensor calibration, timestamp synchronization, software integration, and algorithm evaluation on robotic platforms.

Researcher — Computer Vision and Deep Learning

Ferdowsi University of Mashhad | February 2020–November 2021

- Developed Eynet, an extended YOLO architecture for airport detection in aerial and satellite imagery, using Python, PyTorch, and OpenCV.
- Implemented dataset preprocessing, augmentation, hard-negative mining, and GPU training; evaluated detection in complex remote-sensing scenes, resulting in a first-author journal article.

Graduate Researcher — Nonlinear Control and Intelligent Systems

Ferdowsi University of Mashhad | September 2013–February 2016

- Developed neural-network-based nonlinear model predictive control for parallel robotic systems using sequential quadratic programming and nonlinear system identification.
- Developed a harmonic emotional neural network for system identification and predictive control; research resulted in journal and conference publications.

Teaching Experience

Primary Instructor — EE 407 Introduction to Control Systems

New Mexico State University | Fall 2025

- Independently planned and delivered lectures, assignments, examinations, MATLAB/Simulink activities, and student support.
- Taught dynamic-system modeling, transient and steady-state response, stability analysis, root locus, frequency response, and feedback-controller design.
- Used structured problem solving and simulation to connect mathematical foundations with practical control-system behavior.

Teaching Assistant and Laboratory Developer — EE 475/551 Control Systems Synthesis

New Mexico State University | Spring 2026

- Supported a combined senior undergraduate and graduate course in control-system analysis, synthesis, and implementation.
- Developed three MATLAB/Simulink and Quanser QLabs digital-twin laboratory modules covering transfer-function identification, model validation and robustness analysis, and LQR regulation and tracking.
- Designed laboratory workflows connecting data collection, model fitting, uncertainty analysis, performance evaluation, and controller implementation.

Additional Teaching Assistant Appointments — New Mexico State University

- EE 362 Introduction to Computer Organization — Spring 2026
- ME 210 Electronics and System Engineering — Fall 2024 and Spring 2025
- EE 325 Signals and Systems II — Fall 2023 and Spring 2024
- EE 407 Introduction to Control Systems — Fall 2023

Research Mentoring

- Supported undergraduate students on capstone projects through technical guidance on project development, implementation, and troubleshooting.
- Mentored one M.S. student and one Ph.D. student at New Mexico State University in event-based perception, SLAM, robotic control, software implementation, and experimental design.
- Guided reproducible algorithm evaluation, technical documentation, manuscript development, and research presentations.

Industry and Applied Engineering Experience

Research and Development Engineer

Engarium Planetarium, Ferdowsi University of Mashhad | January 2018–October 2019

- Developed an automated telescope-dome positioning system by integrating motor drives, position sensing, and embedded feedback control to support observatory operation.

Research and Development Engineer

CanDo VR | May 2016–November 2017

- Developed a pneumatic motion-control platform using Raspberry Pi and Unity to synchronize physical motion with a virtual-reality simulation.

Honors and Awards

- Student Travel Support Award, IEEE CCTA — 2025
- RAS Travel Support Award, IEEE CASE — 2025
- Merit-based Enhancement Award, NMSU — Fall 2024 and Spring 2025
- Engineering Research Award, NMSU — Spring 2024
- Graduate Success Scholarship, NMSU — Fall 2022 and Spring 2023

Academic Leadership and Service

- Session Chair, IEEE Conference on Control Technology and Applications — 2025
- Judge, Undergraduate Research and Creative Arts Symposium, NMSU — 2025
- President, Graduate Student Association, NMSU — 2023–2024

Technical Expertise

- **Control and estimation:** Model predictive control, nonlinear and optimal control, distributed control, system identification, Kalman filtering, numerical optimization, and trajectory tracking.
- **Robotics and perception:** Event-based vision, visual odometry, SLAM, place recognition, multisensor fusion, camera calibration, LiDAR processing, and autonomous navigation.
- **Programming and research software:** C++, Python, MATLAB/Simulink, ROS/ROS 2, Linux, Git, OpenCV, PyTorch, and CUDA.
- **Experimental platforms:** Unmanned aircraft systems, mobile robots, neuromorphic cameras, inertial sensors, embedded control systems, and Quanser QLabs.

Publications

Peer-reviewed journal articles

- E. E. Hernández, **H. Mirhajianmoghadam**, E. S. Espinoza, and L. R. Garcia Carrillo, "Predictive Flocking of an Omnidirectional Multi-Agent System for Lattice Formation: Sigma-Norm Design and Comparative Performance Assessment," IEEE Access, vol. 14, pp. 3910-3925, 2026.
- **H. Mirhajianmoghadam** and B. Bolourian Haghighi, "Eynet: Extended YOLO for Airport Detection in Remote Sensing Images," Pattern Analysis and Applications, vol. 29, article 10, 2026.
- **H. Mirhajianmoghadam** and M.-R. Akbarzadeh-T., "Predictive Hierarchical Harmonic Emotional Neuro-Cognitive Control of Nonlinear Systems," Engineering Applications of Artificial Intelligence, vol. 111, article 104781, 2022.

Peer-reviewed conference papers

- **H. Mirhajianmoghadam** and L. R. Garcia Carrillo, "Bringing Loop Closure to Event-Based SLAM: A Place Recognition Approach," Proceedings of the IEEE International Conference on Automation Science and Engineering (CASE), pp. 2124-2129, 2025.
- **H. Mirhajianmoghadam** and L. R. Garcia Carrillo, "EVO-RatSLAM: A Biologically-Inspired SLAM for Event-Based Visual Odometry," Proceedings of the IEEE Conference on Control Technology and Applications (CCTA), pp. 483-488, 2025.
- **H. Mirhajianmoghadam** and L. R. Garcia Carrillo, "Event-Based RatSLAM: Enhancing RatSLAM with Event-Aligned Velocity Estimation and Kalman Filtering for Mapping," Proceedings of the IEEE International Conference on Artificial Intelligence, Robotics and Control (AIRC), pp. 164-168, 2025.
- **H. Mirhajianmoghadam**, N. Grijalva, and L. R. Garcia Carrillo, "Fast Trajectory Tracking for Commercial Off-the-Shelf UAS Using Model Predictive Control: Real-Time Implementation and Performance Analysis," Proceedings of the International Conference on Unmanned Aircraft Systems (ICUAS), pp. 658-665, 2024.
- N. Grijalva, **H. Mirhajianmoghadam**, L. R. Garcia Carrillo, H. Stuckey, and W. Tang, "Low-Cost Monocular Vision-Based Localization and Feedback Control of UAS for Classroom and Education Settings," Proceedings of the International Conference on Unmanned Aircraft Systems (ICUAS), pp. 667-672, 2024.
- **H. Mirhajianmoghadam**, N. Grijalva, and L. R. Garcia Carrillo, "Real-Time Implementation of Model Predictive Control for Fast Trajectory Tracking of a Quad Rotorcraft UAS," Proceedings of the IEEE International Conference on Robotic Computing (IRC), pp. 90-91, 2023.
- **H. Mirhajianmoghadam**, M.-R. Akbarzadeh-T., and E. Lotfi, "A Harmonic Emotional Neural Network for Non-Linear System Identification," Proceedings of the 24th Iranian Conference on Electrical Engineering (ICEE), 2016.

Manuscripts under review

- **H. Mirhajianmoghadam** and L. R. Garcia Carrillo, "eRatSLAM: Event-Based SLAM and Perception for Autonomous Robotic Systems," manuscript under review, 2026.
- H. C. Contreras Aguirre, **H. Mirhajianmoghadam**, O. A. Garcia Alcantara, G. Contreras, and L. R. Garcia Carrillo, "Experiential Learning in Control Systems Education Through Robotics and Digital-Twins," manuscript under review, 2026.

Preprint

- **H. Mirhajianmoghadam** and S. M. Ghasemi, "Efficient Parameter Selection for Scaled Trust-Region Newton Algorithm in Solving Bound-Constrained Nonlinear Systems," arXiv preprint arXiv:2009.04354, 2020.